

COMPANION SLIDES

Chapter 7

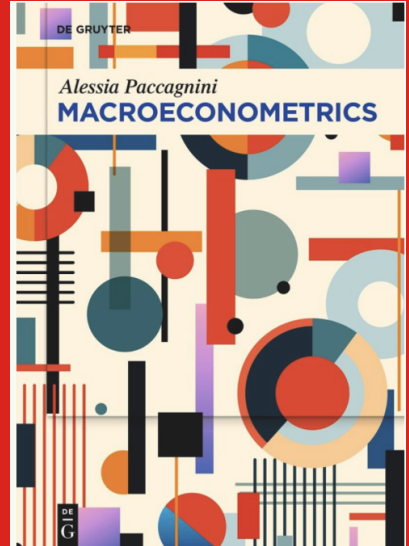
Forecasting

From prediction to forecast evaluation

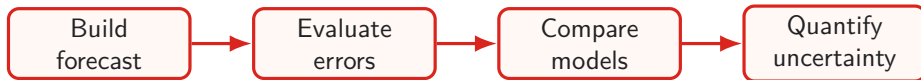
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How do we decide whether a macroeconomic forecast is
accurate, informative and useful?



- ▶ Forecast errors, bias and evaluation schemes
- ▶ Loss functions and accuracy measures
- ▶ Bias–variance tradeoff
- ▶ Direct vs. iterated forecasts
- ▶ Diebold–Mariano and Clark–West tests
- ▶ Forecast instability and Giacomini–Rossi
- ▶ Density forecasts and PIT diagnostics
- ▶ Proper scoring rules and density tests
- ▶ Forecast combination
- ▶ Survey and judgmental forecasts
- ▶ Real-time data and revisions
- ▶ Nowcasting
- ▶ DSGE forecasting
- ▶ Empirical comparison: RW, VAR and BVAR

Point forecast

A forecast $\hat{y}_{t+h|t}$ predicts y_{t+h} using information available at time t .

$$e_{t+h|t} = y_{t+h} - \hat{y}_{t+h|t}.$$

- ▶ $e > 0$: the realization exceeded the forecast.
- ▶ $e < 0$: the forecast was too high.
- ▶ Forecast evaluation studies the behavior of these errors across horizons and origins.

$$\text{Bias} = \mathbb{E}[e_{t+h|t}].$$

Unbiased forecast

$\mathbb{E}[e_{t+h|t}] = 0$. On average, the forecast does not systematically over- or under-predict.

- ▶ Unbiasedness is desirable, but it is not sufficient for accuracy.
- ▶ A forecast can be unbiased and still have very large errors.
- ▶ For overlapping multi-step forecasts, inference on mean errors requires attention to serial correlation.

Scheme	Estimation sample	Useful when
Fixed	unchanged	studying a fixed estimated model
Recursive	expands over time	parameters are relatively stable
Rolling	fixed-length moving window	structural change is a concern

Key takeaway

The evaluation design should mimic the information set and estimation procedure that would have been available in real time.

Squared error

$$L(e) = e^2$$

Large errors receive disproportionate weight.

Absolute error

$$L(e) = |e|$$

More robust to unusually large errors.

Quantile / tick loss

$$L_{\tau}(e) = (\tau - \mathbf{1}\{e < 0\})e$$

Useful for asymmetric decisions and tail forecasts.

The “best” forecast depends on the decision-maker’s loss function.

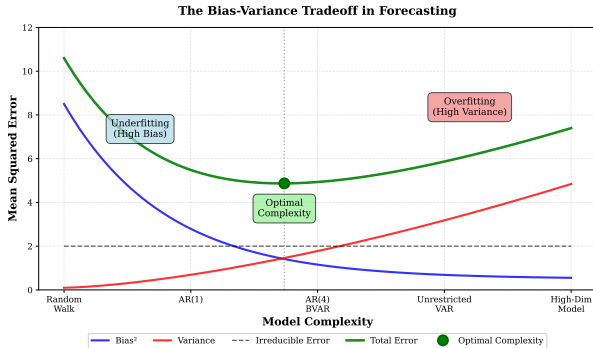
Squared loss: $\hat{y}_{t+h|t}^* = \mathbb{E}[y_{t+h}|\mathcal{F}_t]$,

Absolute loss: $\hat{y}_{t+h|t}^* = \text{median}(y_{t+h}|\mathcal{F}_t)$,

Tick loss: $\hat{y}_{t+h|t}^* = Q_\tau(y_{t+h}|\mathcal{F}_t)$.

Key takeaway

Forecasting is a decision problem: the target statistic changes with the loss function.



Simple models trade lower estimation variance against higher bias; overly complex models can reverse that tradeoff. **Intermediate complexity can minimize total forecast error.**

$$\begin{aligned}\text{RMSE} &= \sqrt{\frac{1}{P} \sum e_{t+h|t}^2}, \\ \text{MAE} &= \frac{1}{P} \sum |e_{t+h|t}|, \\ U_2 &= \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{naive}}}.\end{aligned}$$

- ▶ RMSE emphasizes large mistakes.
- ▶ MAE is easier to interpret and less sensitive to extremes.
- ▶ $U_2 < 1$ indicates improvement over the chosen naive benchmark.

Iterated

Estimate a one-step model and recursively project it forward to horizon h .

Direct

Estimate a separate model specifically for each horizon h .

- ▶ Iteration exploits model structure but compounds misspecification.
- ▶ Direct forecasting is flexible but estimates more horizon-specific parameters.

Loss differential

$$d_t = L(e_{1,t+h|t}) - L(e_{2,t+h|t}).$$

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}.$$

- ▶ Null: equal predictive accuracy.
- ▶ Designed for comparing nonnested forecasting models.
- ▶ Multi-step overlapping errors generally require HAC variance estimation.

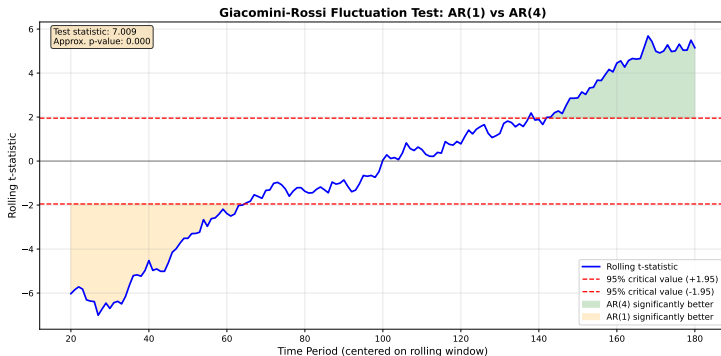
Why DM is not enough

When one model nests another, estimating unnecessary parameters can mechanically inflate the larger model's forecast error under the null.

$$\tilde{d}_t = (e_1^2 - e_2^2) + (\hat{y}_1 - \hat{y}_2)^2.$$

Key takeaway

Use a nested-model adjustment rather than treating the comparison as an ordinary nonnested DM test.



The Giacomini–Rossi

fluctuation framework detects **local changes in relative predictive performance**, especially under structural change or policy-regime shifts.

Question	Tool	Main idea
Nonnested models	Diebold–Mariano	equal average loss
Nested models	Clark–West	correct nesting-induced bias
Instability	Giacomini–Rossi	local relative performance
Conditional performance	Giacomini–White	predictive ability conditional on information
Small samples	Coroneo–Iacone	size refinement under persistence

Point forecast

One number: a mean, median or quantile.

Density forecast

A full predictive distribution $\hat{F}_{t+h|t}(y)$ describing uncertainty, asymmetry and tail risk.

Key takeaway

For policy and risk management, knowing the center of the forecast is often not enough; uncertainty matters.

$$z_{t+h} = \hat{F}_{t+h|t}(y_{t+h}).$$

Calibration benchmark

If the predictive density is correctly specified, PIT values should behave approximately as i.i.d. $U(0, 1)$.

- ▶ U-shaped histogram: predictive density may be too narrow.
- ▶ Hump-shaped histogram: predictive density may be too wide.
- ▶ Skewness can reveal systematic asymmetry or bias.

Log score

$$S_{\log}(f, y) = \log f(y)$$

Rewards probability density placed near the realization; sensitive to assigning very low probability to realized events.

CRPS

$$\text{CRPS}(F, y) = \int (F(x) - \mathbf{1}\{x \geq y\})^2 dx$$

Evaluates the entire CDF and generalizes absolute-error ideas to distributions.

Weighted scoring rules can emphasize economically important regions such as downside tails.

- ▶ The Amisano–Giacomini approach compares average differences in density scores.
- ▶ Quantile calibration tests focus on specific parts of the predictive distribution.
- ▶ Interval forecast evaluation asks whether nominal coverage and observed coverage agree.
- ▶ Density evaluation should assess both **calibration** and **sharpness**.

$$\hat{y}_{c,t+h|t} = \sum_{i=1}^N w_{it} \hat{y}_{i,t+h|t}.$$

- ▶ Different models capture different signals and make different mistakes.
- ▶ Combining can diversify model uncertainty.
- ▶ Equal weights are often surprisingly competitive: the forecast combination puzzle.
- ▶ More elaborate weights can help when relative model performance is persistent and estimable.

Why use professional forecasts?

Human forecasters can incorporate soft information, institutional knowledge and events that are difficult to encode immediately in a statistical model.

- ▶ Surveys provide useful benchmarks.
- ▶ Judgment can improve forecasts when it adds genuine information.
- ▶ Ad hoc adjustments can also introduce bias and inconsistency.
- ▶ Model + survey combinations can exploit complementary information.

- ▶ Macroeconomic releases are revised after their initial publication.
- ▶ A forecaster at date t only has access to the vintage available at t .
- ▶ Evaluating historical forecasts with later revised data can create an unfair information advantage.
- ▶ Revisions can reflect **news** or **noise**, with different implications for forecasting.

Key takeaway

A credible pseudo-real-time exercise should reconstruct the information set that was actually available.

Predicting the present

Nowcasting uses timely high-frequency indicators to estimate current-quarter activity before the official low-frequency release arrives.

$$y_Q = \alpha + \beta' \bar{x}_Q + \varepsilon_Q.$$

- ▶ Bridge equations aggregate monthly indicators into quarterly predictors.
- ▶ Mixed-frequency models avoid discarding timing information.
- ▶ Data releases can update the nowcast sequentially through the quarter.

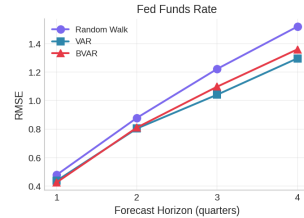
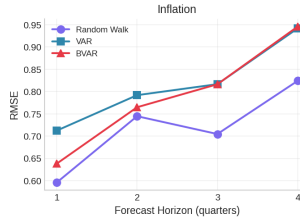
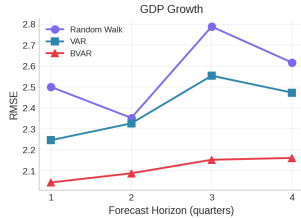
- ▶ Structural restrictions can stabilize estimation and provide economically coherent forecasts.
- ▶ Misspecification can reduce predictive performance relative to flexible reduced-form models.
- ▶ DSGE–VAR methods blend theory-based restrictions with VAR flexibility.
- ▶ Forecast performance is an important empirical discipline for structural models.

Models

Random Walk benchmark, frequentist VAR, and Bayesian VAR with Minnesota prior.

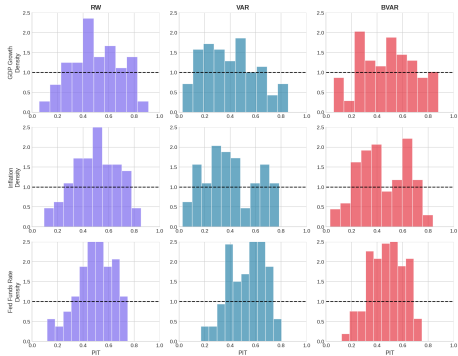
- ▶ Variables: annualized real GDP growth, GDP-deflator inflation and Federal Funds rate.
- ▶ Sample: 1970Q1–2007Q4.
- ▶ Rolling window: initial 60 quarters.
- ▶ Evaluation period begins in 1986Q1.
- ▶ Point and density forecasts are evaluated across horizons.

Point Forecast Accuracy: RMSE by Horizon
(U.S. Data, Evaluation Period 1986-2007)

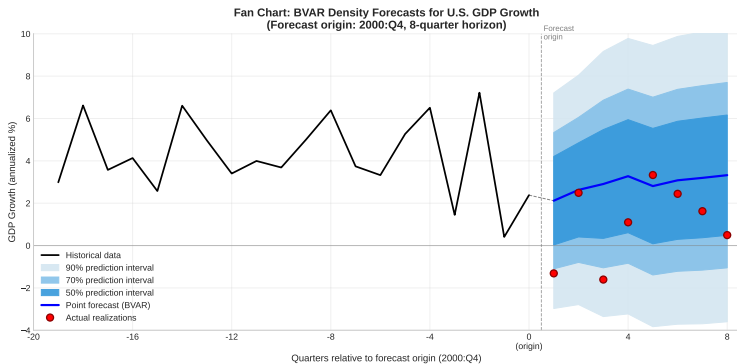


Performance depends on the variable and horizon. **Formal comparison tests** help distinguish sampling variation from meaningful accuracy differences.

Probability Integral Transform (PIT) Histograms
(h=1 quarter ahead, uniform = well-calibrated)



PIT histograms diagnose **calibration**; deviations from uniformity reveal distributional misspecification.



Nested prediction bands
communicate forecast uncertainty, which naturally **widens with the horizon**.

1. Define the target, horizon and information set.
2. Choose realistic benchmarks.
3. Select fixed, recursive or rolling evaluation.
4. Generate genuinely out-of-sample forecasts.
5. Evaluate point accuracy under an appropriate loss.
6. Use the correct formal comparison test.
7. Evaluate density calibration and sharpness when relevant.
8. Check robustness across horizons, vintages and unstable periods.

1. Forecast quality depends on the target, horizon and loss function.
2. Out-of-sample design and real-time information are part of the econometric problem.
3. Statistical significance matters when comparing forecast errors.
4. Density forecasts add information about uncertainty and tail risk beyond point predictions.
5. Simple benchmarks and combinations remain essential, even when using sophisticated macroeconomic models.

- ▶ Compare rolling and recursive forecasts for the same model.
- ▶ Compute RMSE, MAE and Theil's U across multiple horizons.
- ▶ Run DM and Clark–West tests and explain why the appropriate test changes.
- ▶ Construct PIT histograms and diagnose deliberate density misspecification.
- ▶ Combine VAR, BVAR and survey forecasts using equal and estimated weights.
- ▶ Repeat an exercise using real-time vintages rather than revised data.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 7. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your own forecasting models, datasets and horizons.
- ▶ Add coding demonstrations in Python, R or MATLAB.
- ▶ Add survey forecasts, nowcasting indicators and real-time vintages.
- ▶ Add exercises on density forecasting, forecast combinations and instability.

Make them yours.

Factor Models

Extracting information from high-dimensional macroeconomic data

Next question

How can we summarize and exploit large information sets without estimating an unmanageable number of parameters?